PATHOLOGY

1. **Which one of the following is not a characteristic of cancer cell?**
    a) Evasion of apoptosis
    b) Hypersensitivity to growth inhibitors signals
    c) Ability to invasion and metastasis
    d) Development of sustain angiogenesis
2. **excision of small subcutaneous nodule after appendectomy by 28 days microscopic biopsy shows macrophages, collagen, lymphocytes. Which refractile material are seen in nodule and giant cells and surgical complication is?**
    a) Chronic inflammation
    b) suture granuloma
    c) abscess formation
3. **raised nodular scar after 2 months of removing wound suture this is?**
    a) keloid
    b) 2ndary union
    c) orgnization
4. **heart alteration following long untreated HTN?**
    a) atrophy
    b) Hyperplasia
    c) Glycogen storage
    d) Hypertrophy
5. **Predominant cell type in acute inflammation is:**
    a) Neutrophil
    b) monocyte
    c) macrophage
    d) eosinophil
6. **calf muscle affected after long period of immobilization by;**
    a) hypertrophy
    b) atrophy

c) dystrophy

7. eosinophilic staining with loss of nuclei and striation in myocardial tissue?

a) coronary artery thrombosis
b) viral infection
c) chest trauma

8. tumor markers useful in:

a) diagnosis of Ca
b) determining staging
c) assessment of response to treatment and prognosis
d) prediction prognosis

9. Arterial infarction is least likely to be occur in:

a) intestine
b) spleen
c) liver
d) kidney

10.antitumor mechanism:

a) Direct killing by NK Cell
b) phagocytosis by lymphocyte
c) Neutralized by AB
d) Phagocytosis by neutrophils

11.metastatic calcification:

a) In TB
b) Vit D intoxication
c) amyloidosis
d) acute hemorrhagic pancreatitis

12.increased rate of deposition and decrease bone adsorption:

a) increase estrogen
b) decrease parathyroid hormone
c) increase calcitonin

13.Which of the following premalignant lesion:

a) heteropia

b) hamartoma
c) metaplasia
d) hypertrophy

14.in pheochromocytoma which is medullary carcinoma with increased secretion of epinephrine, this led to:

a) decreased VMA
b) decreased heart rate
c) decreased blood pressure
d) sweat calmy skin

15.8 years old female is able to bend her thumb to touch her forearm, skin pull out about 8 cm, she has difficulty in healing of skin cuts. The underlying disease is caused by defect in:

a) collagen
b) factor VIII
c) antitrypsin
d) retinoblastoma protein

16.creatinine kinase is a value test in:

a) muscle disease
b) congestive heart failure
c) liver cirrhosis
d) hemolytic anemia

17.after femur fracture, the patient complains of SOB and took his blood for investigation, they found that is not clot what is the most likely diagnosis:

a) air embolism
b) fat embolism
c) pulmonary thromboembolism

18.generalized edema in heart failure is due to:

a) Lymphatic Obstruction
b) Decreased oncotic pressure
c) Increased hydrostatic pressure
d) acute inflammation

19.in septic shock led to:

a) generalized vasodilatation

b) loss of consciousness
c) hypotension

20. The substance synthesized by fibroblast and connect the cellular integrin to extracellular collagen and to basement membrane

a) fibronectin
b) procollagen
c) dermatan sulphate
d) hyaluronic acid

21. Clotting factor VIII synthesize from:

a) Hepatocyte
b) Kupffer cells
c) Endothelial cells
d) Skin
e) Splenic cells

22. Which of the following is most significant in the diagnosis of Early acute myocardial infarction:

a) ESR
b) ASO titer
c) Troponin
d) LDH
e) CK-BB

23. Von Willebrand factor deficiency mode of inheritance:

a) Autosomal dominant
b) Autosomal recessive
c) X linked dominant
d) X linked recessive
e) Mitochondrial

24. In which of the following conditions does Exudate occur:

a) CHF
b) Decrease protein level
c) Chronic inflammation
d) Renal failure
e) Increase hydrostatic pressure

25. Which of the following bleeding profile consistent with DIC:

a) prolonged PT and APTT increase fibrinogen
b) normal PT and APTT reduced fibrinogen
c) normal PT and prolonged APTT reduced fibrinogen
d) prolonged PT and APTT reduced fibrinogen
e) normal bleeding time

26. Which of the following is true about reversible cell injury?

a) cell membrane fragmentation
b) clumping of chromatin
c) Karyorrhexis
d) Pyknosis
e) No nuclear change

27. Which of the following is responsible for the regression of the Uterus size after deliver:

a) Endoplasmic reticulum
b) Golgi apparatus
c) Nucleus
d) Mitochondria
e) Lysosomes

28. Which one of the following is a metastatic calcification:

a) Schistosomiasis
b) Pulmonary tuberculosis
c) Bening prostatic hyperplasia
d) sarcoidosis

29. Which of the following is true about DKA:

a) omit of insulin dose
b) hypokalemia is feature
c) acidosis occurs due to hyperglycemia
d) Hypoosmolality

30. Which of the following immunoglobulin is the largest:

a) IgM
b) IgG
c) IgD
d) IgE
e) IgA

31. Patient with chronic renal failure has normocytic normochromic Anemia what is the best treatment to correct anemia:

a) erythropoietin
b) ferrous sulphate
c) folic acid
d) vit B6
e) vit B12

32. Which of the following mediator is likely to cause pain and increase in vascular permeability?

a) bradykinin
b) prostaglandins
c) Il-1
d) Leuckotrien

33. Early sign detected in diabetic nephropathy is?

a) Pyuria
b) microalbuminuria
c) high urea
d) high creatinine
e) persistent glucosuria

34. After delivering a baby a sample was taken from the umbilical Cord and electrophoresis done. We found abundant Immunoglobulin In the sample. This is:

a) IgM
b) IgG
c) IgE
d) IgA
e) IgD

35. Most specific Enzyme that increases due to liver damage is:

a) GoD
b) GPT
c) ALP
d) Bilirubin
e) Serine hydroxyamelase

36. Nephrotic syndrome:

a) High cholesterol levels
b) Oliguria is a feature
c) mild proteinuria
d) high blood albumin
e) no swelling

37.All of the following are examples of physiological cells Death(apoptosis) except:

a) Thymic aplasia
b) Menstruation
c) RBCs at spleen
d) development of digits
e) Viral hepatitis

38.Lymphedema caused by:

a) congestive heart failure
b) accumulation of lymph extracellularly
c) secondary from others organ
d) increase hydrostatic pressure

39.Which of the following occur with chronic renal failure:

a) decrease serum urea
b) hypokalemia
c) hypocalcemia
d) decrease creatinine clearance
e) hyponatremia

40.Benefit of acute inflammation include:

a) fibrosis
b) vasoconstriction
c) initiate immunity
d) abscess formation
e) lymphocytosis

41.Following true about necrosis

a) Gangrene necrosis is gangrene + infarction
b) Fibrinoid necrosis come with malignant HTN
c) Brain necrosis is coagulative necrosis

42.Which type of antibodies will associate in blood cell coagulation

a) IgE
b) IgM
c) IgA
d) IgG

43.P53

a) Mutated in tumor cell
b) Oncogene
c) Used in cell replication
d) DNA repair

44.All of the following are true about osteoporosis except

a) Can be primary or secondary
b) Increased osteoblast activity
c) occur in vitamin D deficiency
d) occur in vitamin C deficiency

45.Vaso occlusive crisis in patient with sickle cell anemia is

a) Hand and foot syndrome
b) Auto splenectomy

46.Drug induce asthma by

a) Type I hypersensitivity reaction
b) Induce immune reaction
c) act as hapten

47.After a long period of corticosteroid therapy, wound healing is delayed due to:

a) Collagen damage
b) Fibroblast elaboration

48.pyogenic bacterial infection and ischemic brain infarcts most characteristically produce which one of the listed types of necrosis

a) Caseous necrosis
b) Coagulation
c) Fat necrosis
d) liquefactive necrosis

49.Anasarca occurs due to

a) Complicated pregnancy induce hypertension
b) surgical removal of lymph node
c) Nephrotic syndrome

50.Breakdown of 1gm Haemoglobin produces:

a) 20 mg of bilirubin
b) 35 mg of bilirubin
c) 50 mg of bilirubin
d) 70 mg of bilirubin

51.Unconjugated bilirubin in serum is soluble in:

a) Water
b) Alkalis
c) Acids
d) Methanal

52.The active transport system for hepatic uptake of bilirubin is congenitally defective in:

a) Gilbert's disease
b) Crigler-Najjar syndrome type 1
c) Rotor's syndrome
d) Crigler-Najjar syndrome type 2

53.Excretion of conjugation bilirubin from liver cells into biliary canaliculi is defective in:

a) Gilbert's disease
b) Crigler-Najjar syndrome
c) Lucey-Driscoll syndrome
d) Rotor's syndrome

54.In iron deficiency anemia:

a) mobilization of hemosiderin reserves.
b) activation of ferritin phage
c) elevated intestine apo ferritin

55.Red infarction develops in:

a) Spleen
b) Liver
c) Kidney
d) Intestine

56.grading of tumor is best described as:

a) degree of differentiation
b) size of tumor

c) lymph nodes
d) metastasis

57.pure hypertrophy is seen in:

a) breast of pregnant women
b) uterus of pregnant women
c) muscle of athletes

58.How many atoms of Iron are found in one molecule of Hemoglobin:

a) 2
b) 4
c) 6
d) 10

59.Healing by first intention:

a) the surface of wound is rapidly covered by epithelium
b) excessive granulation tissue
c) wound will cover by fibroblast

60.In cancerous cell

a) loss of architecture

61.Definitive sign of necrosis:

a) decrease in PH
b) ribosomal detachment
c) karyolysis
d) cell swelling

62.After treatment of myocardial infarction, the injury occurs instead of restore blood supply by:

a. free radical

63.Keloid due to:

a) increase collagen deposition
b) increase proliferation of fibroblast
c) neoplasia of fibroblast

64. 5% of total body iron is found in which form:

a) hemosiderin
b) transferrin
c) myoglobin

d) cytochrome

65.Complement based agglutination reaction is known as:

a) heme agglutination
b) complement fixation
c) conglutination
d) Schultz dale phenomenon

66.Wound healing by regeneration invariable in:

a) liver cells
b) renal tubules
c) neuron
d) stem cells

67.Most of iron in the body is present in:

a) RBCs
b) Liver
c) Plasma
d) Spleen

68.Epithelium cell tumor:

a) Papilloma
b) Cystadenoma
c) Choriocarcinoma
d) Nephroblastoma

69.Which of the following activates carcinogenic materials:

70.Pre-cancerous screening is possible in:

a) ovary
b) cervix
c) uterus

71.Transformation Zone:

a) glandular to squamous metaplasia
b) glandular to squamous dysplasia
c) squamous to glandular dysplasia
d) squamous to glandular metaplasia

72.Choriocarcinoma metastasis, it has a predominantly spread by which route?

a) direct invasion
b) haematogenous
c) lymphatic
d) transcolomic

73.From which organ/system does the Kruckenberg tumor of the ovaries originate?

a) Lung
b) Brain
c) Uterus
d) GIT

74.Cell of chronic inflammation is:

a) macrophages with giant cell
b) polymorph nuclear
c) monocyte
d) plasma cell

75.Which of the following is true about exudate:

a) occurs in heart failure
b) due to chronic inflammation
c) used in cytology cultures
d) has low specific gravity

76.Free radical:

a) destruction of protein sh group
b) causes apoptosis

77.Endothelium adhesion by:

a) selectin
b) integrin
c) mucin
d) fibronectin

78.Active immunity is induced by

a) Infection
b) Placental transfer of antibodies
c) Injection of antibodies
d) Injection of gamma- globulins

79.Irreversible cell injury is characterized by:

a) Cell swelling
b) Lysosomal ruptured

c) Cell membrane defects
d) Nuclear chromatin clumping

80.Chronic inflammation is

a) always preceded by acute inflammation
b) characterized by hyperemia, oedema and leukocyte infiltration
c) most frequently results in resolution
d) the factors underlying monocyte infiltration are the same as for acute inflammation

81. Chemotaxis is

a) Coating of antigen by antibodies
b) vasodilation after chemotherapeutic agent
c) Migration of cell toward specific site
d) Increase blood flow in the site of inflammation

82.Histamine cause vasodilation and increase permeability on:

a) Arteriole
b) Veniole
c) Capillaries
d) Post capillary venioles

83. 28-year-old female presented to the hospital with pelvic pain, when endometrial biopsy was taken, there is a feature of chronic endometritis, what type of cell would be there?

a) Neutrophils
b) lymphocytes
c) plasma cells
d) Giant cell

84.Antigens and haptens have the following similarity:

a) They have high molecular weights
b) They can elicit immune response by themselves
c) They can elicit an immune response only in association with some other large molecule
d) Once an immune response develops, free antigen and free hapten can be recognized by the antibody

85.What is the first sign of bone healing:

a) Lamellar bone
b) Woven bone
c) Collagen
d) Inflammation

86.In Acute pancreatitis:

a) Alcohol causes direct damage to islet cells
b) Cause necrosis to intraperitoneal fats

c) Caused by trauma in 30%

87.Hepatorenal syndrome:

a) increase renal blood flow
b) loss ability to concentrate urine
c) oliguria characteristic
d) Urine is hyperosmolar
e) High sodium in urine

88.Metaplasia:

a) Irreversible
b) Barrett's oesophagus an example
c) 70% in twins

89.type 1 diabetes pathogenesis:

a) autoimmune insulitis
b) Led to glucokinase
c) Due to insulin insensitivity

90.Liver cirrhosis:

a) Fibrosis confined to delicate band around central veins
b) Ito cells are responsible for excess production of fibrosis
c) Left lobe of liver is more common affected

91.Bronchial asthma:

a) Extrinsic asthma non selection immune mediated
b) eosinophils and neutrophils had roll

92.Type 2 hypersensitivity:

a) Serum sickness is example
b) In transfusion reaction
c) Cell mediated complex

93.In haemolytic jaundice:

a) Unconjugated bilirubin bound tightly to albumin
b) Conjugated bilirubin bound tightly to albumin
c) Bilirubin concentration in urine increases
d) Unconjugated bilirubin increases in urine
e) Cause kernicterus in adults

94.Acute tubular necrosis:

a) cause hyperkalaemia not hypokalaemia
b) non oliguric better recovery
c) ischemic cortical cell
d) 80% anuria

95.Pyelonephritis:

a) 85% caused by gram negative bacteria
b) obstruction and spread of infection
c) Cause abscess of kidney

96.Dystrophic calcification in heart valves:

a) Not cause dysfunction
b) rarely occur in mitochondria
c) cardiac valves not involve
d) deposit of crystal calcium

97.Compensated hypertrophic cardiomyopathy:

a) High capillary/myocyte ratio
b) increase fibrous
c) increase vascular density
d) hyperplasia

98.Question about smooth muscle atrophy…

﴿وَمَا تَوْفِيقِي إِلَّا بِاللَّهِ ۚ عَلَيْهِ تَوَكَّلْتُ وَإِلَيْهِ أُنِيبُ﴾